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Course 608

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Research Topic: Predictors of Diagnostic Delay and Their Correlation With Disease Severity (DAS28) in Pakistani Rheumatoid Arthritis Patients.

Title: A Retrospective Cohort Study of the predictors of diagnostic delay and their correlation with Disease Severity (DAS28) in Pakistani Rheumatoid Arthritis Patients

Research Problem:

Rheumatoid Arthritis (RA), a progressive autoimmune disorder, is a growing burden in Pakistan as the incident and prevalent cases increased markedly from 8,606 to 19,454 and from 134,628 to 318,963, respectively between 1990 and 2021 [1]. Early initiation of disease-modifying anti-rheumatic drug (DMARD) therapy within the first 12 weeks of symptom onset, widely recognized as the critical "window of opportunity", is vital for preventing irreversible joint erosions and halting long-term structural disability [2]. However, translating this global clinical mandate into local practice faces severe hurdles within Pakistan's healthcare ecosystem. Research has revealed that only 22% (52 out of 235) of Pakistani RA patients managed to reach a rheumatologist within this crucial 12-week timeframe [3]. This stark contrast emphasises that the vast majority of local patients miss the therapeutic window entirely, establishing a clear need to investigate the specific referral barriers and structural delays driving this gap in the indigenous population. The systemic ways behind these prolonged delays come from a mix of primary care referrals first visits that are not, with specialists and a wide use of alternative medicine.Naeem et al. (2021) demonstrated that only 12.7% of Pakistani RA patients initially consult a rheumatologist, with the vast majority first presenting to orthopedic surgeons (39.2%) or general practitioners (26.5%) [4]. This failure of early triage leads to a median diagnostic lag (Lag-3) of 12 months, where visiting more than three doctors prior to diagnosis strongly correlates with extended delay (r = 0.6) [6]. This pattern is echoed regionally by Khan et al. (2025) in Quetta, who reported a mean diagnostic delay of 17.8 ± 27.9 months and identified female gender as an independent predictor of delayed diagnosis (p = 0.014) [5]. Beyond formal healthcare pathways, health-seeking behavior in Pakistan is heavily influenced by alternative medicine; Akram et al. (2020) note that 60% to 90% of chronic arthritis patients utilize traditional therapies [6]. The formal clinical integration of these alternative pathways is supported by Wagan et al. (2022), who successfully documented "Hakeem medications" as a discrete, chart-recorded variable in local RA cohorts [7]. Importantly, despite these multi-factorial delays and their documented correlation with higher disease activity and functional disability, retrospective chart evaluation remains a highly viable methodology. Demonstrating this feasibility, Butt et al. (2024) successfully extracted baseline DAS28 scores (mean 4.5 ± 1.4) from medical records in treatment-naïve Pakistani patients, establishing that clinical disease activity indices can be reliably captured from local hospital registries to evaluate the impact of diagnostic lag [8]. This study addresses this gap by evaluating DAS28 directly at diagnosis in treatment-naïve patients using an adjusted multivariable model to isolate the independent predictors of diagnostic delay.

Existing Literature:

Global research demonstrates that diagnostic delays in rheumatoid arthritis (RA) exacerbate structural joint damage and reduce clinical remission rates [2]. While Western studies attribute delays to specialist shortages and seronegative presentations, South Asian populations face distinct healthcare barriers [9]. Regional epidemiological data from Pakistan confirm an increasing RA burden alongside persistent referral bottlenecks [1,4]. Patients frequently present first to primary care physicians or non-rheumatology specialists, resulting in extended median diagnostic lags [4,3]. Local evidence shows that non-polyarticular onset patterns and non-hand presentations significantly contribute to diagnostic delays (e.g., mean delay of 17.8 months in Pakistani cohorts) [5]. However, limited empirical data directly links specific serological markers (anti-CCP seropositivity vs. seronegativity) and socioeconomic factors to baseline disease activity (DAS28) and functional disability (HAQ-DI) at initial diagnosis. Prior regional evaluations have largely relied on unadjusted bivariate correlations in treated cohorts [4]. This study addresses these gaps through an adjusted multivariable analysis in treatment-naïve patients within the Pakistani healthcare context.

Research Question:

To what extent is diagnostic delay independently associated with baseline disease severity (DAS28) at the time of diagnosis in treatment-naïve Pakistani Rheumatoid Arthritis (RA) patients, considering adjustments for clinical and socio-demographic factors such as age, gender, education level, serostatus, and initial consultation provider?

Methodology:

This study adopts a mixed-methods approach to ensure a comprehensive understanding of both measurable clinical outcomes and underlying healthcare-seeking experiences:

Quantitative Component:

- **Study Design & Data Tool:** A retrospective cohort design will extract secondary clinical data from medical records of treatment-naïve Rheumatoid Arthritis (RA) patients across tertiary care rheumatology outpatient clinics using a REDCap data entry tool.
- **Measured Variables:** The REDCap tool will capture baseline disease severity (DAS28-ESR at initial diagnosis), total diagnostic delay (defined per Naeem et al.'s framework as Lag-3: duration in months from symptom onset to confirmed diagnosis), binary delay thresholds (>12 weeks vs. ≤12 weeks), serological profiles (anti-CCP and RF titers), presenting joint patterns, and healthcare utilization history (first provider consulted, history of misdiagnosis, and non-allopathic "Hakeem" drug use).
- **Sample Size & Sampling Procedure:** A sample size of N = 250 adult patients (aged 18 years and above) meeting the 2010 ACR/EULAR classification criteria will be selected via systematic random sampling from hospital registries (selecting every nth eligible consecutive chart recorded during the study window). This sample size was calculated to achieve >80% statistical power (alpha = 0.05) for detecting moderate effect sizes ($f^2 = 0.15$) in multivariable linear regression, balancing statistical robustness against multi-center feasibility.
- **Statistical Analysis:** Multivariable linear regression will evaluate the independent association between total diagnostic delay (Lag-3, in months) and baseline DAS28-ESR score, explicitly controlling for age, sex, monthly household income, formal education level, serostatus, and initial contact provider. Multivariable logistic regression will identify independent predictors of experiencing diagnostic delay exceeding the 12-week therapeutic window.

Qualitative Component:

- **In-Depth Interviews (IDIs):** IDIs will be conducted with 20 treatment-naïve RA patients, 5 first-contact non-specialist healthcare providers (general practitioners and orthopedic surgeons), and 3 consulting rheumatologists to explore fundamental referral bottlenecks and decision making pathways.
- **Focus Group Discussions (FGDs):** FGDs will be held with patients and primary caregivers to examine socio cultural health beliefs, perceptions of early joint symptoms, and motivations driving the initial reliance on alternative therapies ("Hakeem medications").
- **Qualitative Analysis:** Audio recordings will be transcribed, translated, and analyzed thematically through induction with NVivo software to identify recurring systemic and behavioral themes.

Ethical Considerations:

- **Institutional Review:** Formal ethical approval will be secured from the Institutional Review Board (IRB) of each participating tertiary hospital prior to chart retrieval and recruitment.
- **Informed Consent:** Written informed consent will be obtained from all interviewed patients, healthcare providers, and FGD participants.
- **Data Anonymity:** All secondary clinical data and qualitative transcripts will be fully anonymized and assigned unique alphanumeric identifiers to maintain strict confidentiality in accordance with medical data privacy protocols.

Research Topic Justification:

While regional literature has established the high prevalence of diagnostic delay in Pakistani Rheumatoid Arthritis (RA) cohorts, existing studies, most notably Naeem et al. (2021)—remain limited to unadjusted bivariate evaluations in patients with longstanding, treated disease [4]. This study delivers a primary methodological innovation by generating the first adjusted multivariable assessment of baseline disease severity (DAS28-ESR) specifically at the point of diagnosis in treatment-naïve patients.The project contextually highlights the distinct, socio-culturally rooted referral challenges within Pakistan's healthcare system, specifically measuring the effect of initial visits to non-rheumatology specialists (such as orthopedic surgeons and general practitioners) and the prevalent dependence on non-allopathic "Hakeem" remedies noted by Akram et al. (2020) and Wagan et al. (2022) [6,7]. By identifying the independent factors that lead to prolonged diagnostic delays past the essential 12-week period, these empirical results offer practical information to create focused educational programs for non-specialist first responders and develop uniform primary care triage protocols within local outpatient networks.

Quality of Writing:

This proposal has been written and proofread with attention to clarity, clinical precision, and structural coherence for an audience of academic and clinical reviewers.

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